

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

by

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4 Implementation and Empirical Results

Chapter 3 described how the scores are built, which checks are applied, how uncertainty is quantified and how runtime is measured. This chapter runs that design on Arbitrum One as a single case study: one chain and one observation window, one matched cohort for every same-window comparison, a spender cohort and a trader cohort for the temporal holdouts, and one expanded address pool for node-count sensitivity, with a registered replication on June to August 2026 at the end of Section 4.6. Every table is generated from the evaluation summaries of Section 3.13, and each Kendall τ comes with its confidence interval. Each result is reported for all the scores together, each PageRank is read against the raw degree it smooths, and the paired differences between EndorseRank and AWP are in Appendix 8.5.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because one of its margin-trading protocols, GMX V2, supplies the wallet seed set. The observation window, 1 December 2025 to 31 May 2026, spans several market regimes, and within it the time weight of an event never falls below about half of the weight of a recent one (Figure 3.3).

4.1.1 Cohorts

Three address sets are used.

- Matched cohort ($n = 5,521$). The GMX V2 wallets with at least three closes in the window, fixed from GMX data before any approval or transfer was extracted (Section 3.2). Membership does not depend on either graph: no allowance edge touches 966 cohort wallets, no

transfer edge touches 381, and 370 are touched by neither; a wallet absent from a graph scores zero under that method. Every same-window alignment, difference test and benchmark is computed on this cohort.

- Holdout spender cohort ($n = 1,335$). Every spender that holds at least one positive latest allowance at t_1 (28 February 2026, 23:59:59 UTC) in the extracted approval data. Most are contracts. Read from an Arbitrum node at the end of September 2026 (block 510,408,687), 1,174 of the 1,335 addresses held contract code and 161 were externally owned accounts, one of which delegates to contract code under EIP-7702. All 100 spenders that EndorseRank ranks highest are contracts, as are 221 of the 222 that gained a new approval pair from March to May. A BigQuery check of the scoring window agrees where it can decide, since every spender that emitted an event log holds code. Of the 686 spenders that neither emitted a log nor sent a transaction in the window, 530 are contracts, among them routers whose events other contracts emit, such as the GMX exchange router named in the pipeline configuration, and 156 are externally owned accounts. The scores of these wallets are tested against owner–spender approval pairs and transfer senders first seen from 1 March to 31 May 2026 (Section 3.7).
- Expanded wallet pool ($N = 99,080$). The matched cohort plus 93,559 addresses sampled from the chain’s monthly ERC-20 activity, used only for the node-count sensitivity of runtime and the sensitivity to sample definition (Section 3.12.1). The added addresses bring no edges, so the full pool’s solver graph is the cohort graph.

Table 4.1 lists the graph sizes. On the same wallets the endorsement graph has about one-ninth as many edges as the transfer graph, because EndorseRank keeps one latest allowance per token, owner and spender, whereas AWP adds a term for every transfer. The sampling is the same for both, so the differences between the scores trace back to their edges and restarts.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two

edge lists node by node (Equation (3.9)). Each run writes its scores, per-proxy statistics, bootstrap intervals and timings to one file. The tables of this chapter come from runs made on 2 October 2026, except those of the registered replication, evaluated on 1 October; Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C1)

Claim C1 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the matched cohort, the PageRank solve time and its standard deviation over the five timed repeats, the peak memory of the warm-up solve, the number of iterations to convergence and the size of the evaluation subgraph.

Table 4.1: Computational benchmark on the matched wallet cohort ($n = 5,521$; AWP/EndorseRank runtime ratio $9.2\times$). Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs. $|V|$, $|E|$ = nodes and edges of each method’s evaluation subgraph. C-PR walks on the union of the allowance layer and the transfer layer, so $|E|$ is the sum of the two edge sets; S-PR runs on the transfer layer and its runtime includes the solve of the allowance layer that supplies its restart vector.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$ V $	$ E $
EndorseRank	0.547	0.014	4.6	37.0	6,442	14,727
AWP	5.029	0.062	38.2	91.0	30,791	137,087
C-PR ($\lambda = 0.5$)	5.667	0.091	41.3	76.0	31,242	151,814
S-PR	5.600	0.020	39.1	145.0	30,791	137,087

Most of the timed call is spent before the first iteration, when the sparse matrix is assembled from the edge list in a per-edge Python loop. A separate profiling run of the same call on the same edge lists attributes about 94% of the wall time to matrix assembly (0.52 of 0.55 s for EndorseRank, 4.84 of 5.17 s for AWP). The power iterations take about 1%: about 6 ms for EndorseRank’s 37 iterations and 67 ms for AWP’s 91. Node indexing takes most of the rest, and forming AWP’s restart vector about 11 ms. Assembly and each iteration are both linear in $|E|$ (Appendix 8.3.3), so the measured ratio of $9.2\times$ tracks the edge ratio of $9.3\times$. EndorseRank also converges in fewer

iterations. The allowance graph has no cycle apart from one self-loop, and 99.0% of its edges end at spenders without out-edges, so every path stops after a step and the mass returns through the restart distribution; the transfer graph has a strongly connected core of 9,278 addresses, and 28.5% of its edges are reciprocated, so the walk circulates for longer. That difference is real, but it adds little to the wall time. Peak memory likewise grows with $|E|$, and run-to-run noise is about two orders of magnitude smaller than the gap between the methods (standard deviations of 0.014 s and 0.062 s).

The two hybrid rows show what the speed result covers and where it stops. C-PR walks on the union of both edge sets (151,814 edges, 31,242 nodes) and needs 5.67 s, longer than AWP. It converges in fewer iterations (76 against 91), but it has more edges to assemble, and assembly dominates the call. S-PR runs on the transfer layer and takes 5.60 s. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector: the 145 iterations are 37 for that solve plus 108 for the seeded transfer walk. The low cost thus belongs to EndorseRank alone: any algorithm that queries the transfer layer, on its own or combined with allowance information, pays the cost of that query.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one, so the runtime has to be read against the alignment and holdout results that follow.

4.2.1 Node-count sensitivity on the expanded pool

Deterministic SHA256 subsampling of the expanded pool yields three increasing node counts. Table 4.2 reports the runtime at each of them, together with the nodes and edges of every algorithm’s subgraph at that stage. The in-cohort stages ($n \leq 5,521$) are in Appendix 8.1 (Table 8.1).

Two features of Table 4.2 follow from Section 3.12.1. The full-pool row is identical to the matched-cohort row of Table 4.1 and to the $d = 0.85$ row of Table 4.6, by design: no logs were parsed for the extra addresses, so the subgraph induced at $N = 99,080$ is the cohort subgraph, and each distinct

Table 4.2: Runtime versus evaluation-set size on the expanded wallet pool ($N = 99,080$; SHA256 subsampling seed benchmark-tier2-v1). Each stage keeps the edges that touch at least one sampled wallet, so $|V|$ and $|E|$ are reported per stage; the pool adds addresses but no additional extracted edges beyond the matched-cohort subgraph, hence the full-pool row reproduces the matched-cohort graph. Speedup = AWP/ER runtime ratio within the same row. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
10,000	0.075	1.078	$14.4\times$	1,296	2,055	8,489	26,732
50,000	0.342	3.414	$10.0\times$	4,440	9,190	19,990	92,887
99,080	0.547	5.029	$9.2\times$	6,442	14,727	30,791	137,087

graph is timed once per session. And $|E|$ grows with n within each scaling table but not across the two samplings: the pool stage with $n = 10,000$ has fewer EndorseRank edges (2,055) than the in-cohort stage with $n = 1,000$ (2,584, Table 8.1), because only the cohort wallets, about 6% of the pool, had their logs extracted. What the table supports is a bounded claim. Across stages whose edge counts differ by a factor of five to seven, AWP takes 8.6 to 14.4 times as long as EndorseRank, and EndorseRank stays at or below 0.55 s; the largest ratio belongs to the smallest pool stage, where AWP needs 97 iterations and EndorseRank 32. How either algorithm scales as edges grow with the number of addresses is not shown, because the second-stage extraction that was to supply the extra edges did not finish. Figure 4.1 plots the in-cohort and pool stages on one log axis.

4.3 Same-window alignment (Claim C2)

Claim C2 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap confidence interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. Apart from the secondary rule for C-PR fixed on 14 September (Section 3.7), none of the comparisons below was fixed in advance (Chapter 3).

The scores divide by the edges they walk on. Those that walk on transfers, AWP, C-PR at every λ

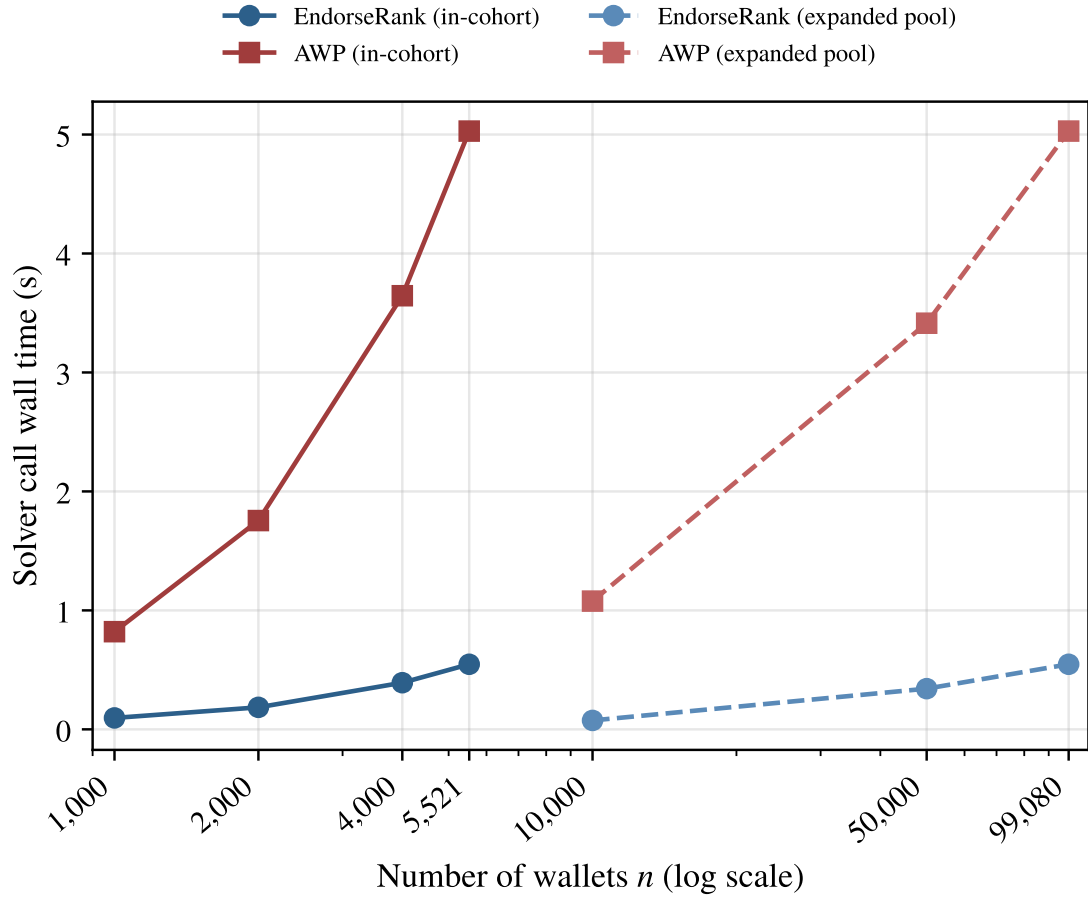


Figure 4.1: Wall-clock time of one solver call against the number of sampled wallets (log x -axis). Solid lines: in-cohort stages (Table 8.1). Dashed lines: expanded-pool stages (Table 4.2), which add no extracted edges, so the rightmost pool point is the full matched cohort. Runtime follows the edge count of each stage and not n .

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this thesis (matched cohort, $n = 5,521$). The scores are EndorseRank and AWP, the coupled operator at five values of λ and the endorsement-seeded walk; $\lambda = 1$ is the allowance layer walked alone (latest allowances in raw amounts) and $\lambda = 0$ the transfer layer walked alone (raw amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.330 [0.311, 0.350]	0.375 [0.342, 0.401]	0.231 [0.212, 0.250]
AWP	0.612 [0.602, 0.622]	-0.024 [-0.043, -0.002]	0.278 [0.264, 0.292]
C-PR ($\lambda = 1$)	0.333 [0.314, 0.353]	0.355 [0.322, 0.384]	0.233 [0.214, 0.253]
C-PR ($\lambda = 0$)	0.534 [0.522, 0.547]	-0.030 [-0.045, -0.017]	0.286 [0.274, 0.300]
C-PR ($\lambda = 0.25$)	0.531 [0.520, 0.545]	-0.009 [-0.020, 0.004]	0.288 [0.276, 0.302]
C-PR ($\lambda = 0.5$)	0.527 [0.515, 0.540]	-0.001 [-0.013, 0.013]	0.290 [0.279, 0.304]
C-PR ($\lambda = 0.75$)	0.519 [0.507, 0.533]	0.006 [-0.007, 0.020]	0.292 [0.281, 0.306]
S-PR	0.494 [0.481, 0.508]	0.116 [0.097, 0.134]	0.220 [0.207, 0.235]

below 1 and S-PR, reach 0.494 to 0.612 on the transfer family and stay near zero on the allowance family, where only S-PR, whose restarts come from the allowance walk, reaches 0.116. The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.375 and 0.355 on the allowance family and about 0.33 on the transfer family. On the Sybil-stability family every score lies between 0.220 and 0.292. Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and in-value from the transfers and in-approve degree and value from the latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's own edge type is therefore an intended-construct check and cannot count as an external test. The holdout of Section 4.6 asks the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP's largest coefficient is with transfer in-degree ($\tau = 0.728$); with in-value it is much lower (0.495). That is what its weights lead one to expect: V counts every transfer of ten base units or more about once, whatever its amount, so AWP follows the number of incoming transfers more closely than their summed value. It does not collapse to the in-degree either. Propagation, time weights and

restarts at active senders shift a measurable share of pairwise orderings, and out of window AWP ranks new senders below the in-degree (Section 4.6). On the Sybil-stability family AWP follows tenure and active months, and it is clearly negative on the inbound-counterparty ratio (-0.228). Its edge weights count transfers, so a wallet that receives many transfers from a few senders ranks high, and that is the pattern the ratio penalizes.

EndorseRank's coefficients are decided by its ties (Section 4.4). Only 131 cohort wallets lie above the common value of the wallets that receive no approval, both allowance proxies are positive for the same 131 wallets, and EndorseRank lifts all of them above its common value. Its two allowance coefficients (0.376 and 0.373) therefore mostly measure the same split and sit at the ceiling that the ties allow (0.377 and 0.378). Against a proxy without ties the ceiling is about 0.57, so EndorseRank's coefficients on different families are not on one scale. Its transfer coefficient (0.330) mainly records that wallets in the allowance graph also receive more transfers than wallets outside it, and both coefficients change sharply when the wallets outside the graph are kept as isolated nodes (Section 4.4). C-PR ($\lambda = 1$) walks the same allowance graph with raw amounts, also restarts uniformly, and stays within 0.02 of EndorseRank on every family.

The same-window trading families of the research proposal, liquidation, inverse risk and GMX trading success (Section 2.8), are archived in Table 8.2. Every score aligns weakly with them. No family mean exceeds 0.24, and once two inverse-risk proxies are oriented correctly (Appendix 8.1.5), every score is negative on the inverse-risk family. The summed loss and the worst close get larger the more a wallet trades, and on the non-loss close rate, a share that does not, the coefficients of the scores lie between -0.031 and 0.024 .

Claim C2 is supported in a narrow sense. Every score that walks on one edge type agrees with the proxies of that edge type, with an interval well clear of zero, and the mixtures follow the denser of their two layers. The agreement is partly built in, and for EndorseRank it is set by the rule that scores a wallet outside the graph zero.

4.3.1 The coupled operator (RQ5)

Two hybrids were defined in Section 3.6.3. The coupled operator C-PR mixes the raw-amount allowance and transfer layers in each wallet’s transition row with weight λ , and the endorsement-seeded walk S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.4 gives their paired contrasts in this window: first the eight of Section 3.10 against whichever layer leads each family, then four against EndorseRank and AWP. RQ5 is decided by the out-of-window tests in Section 4.6. Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

Table 4.4: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the layer that leads each family walked alone, the allowance layer (C-PR ($\lambda = 1$)) or the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 5,521$). Family-mean τ ; the same wallet resamples are used for a and b . The contrasts with the endpoints are a stricter check beside the secondary criterion fixed on 14 September 2026 (Section 3.7), which compares C-PR’s point estimate with the leading layer’s interval in Table 4.3. An interval below 0 means a is significantly below b ; one above 0 means it exceeds it. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
<i>Against the two layers walked alone</i>					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.527	0.534	-0.007	[-0.009, -0.005]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	-0.001	0.355	-0.356	[-0.388, -0.323]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.290	0.286	+0.004	[0.003, 0.006]	1.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.290	0.233	+0.057	[0.043, 0.069]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.494	0.534	-0.040	[-0.049, -0.030]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.116	0.355	-0.239	[-0.263, -0.215]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.220	0.286	-0.065	[-0.072, -0.058]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.220	0.233	-0.013	[-0.025, -0.003]	0.007
<i>Against EndorseRank and AWP</i>					
C-PR transfer minus AWP transfer	0.527	0.612	-0.085	[-0.095, -0.075]	0.000
C-PR allowance minus EndorseRank allowance	-0.001	0.375	-0.375	[-0.404, -0.344]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.290	0.278	+0.012	[0.005, 0.020]	1.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.290	0.231	+0.059	[0.045, 0.072]	1.000

C-PR behaves like its transfer layer with a small allowance-layer correction. Its transfer-family coefficient falls from 0.531 at $\lambda = 0.25$ to 0.519 at $\lambda = 0.75$, always below C-PR ($\lambda = 0$) at 0.534 and within 0.015 of it. At the primary setting the paired difference to that walk is -0.007 (interval

$[-0.009, -0.005]$), small but clear of zero. AWP ranks the transfer family better than either: C-PR lies 0.085 below it ($[-0.095, -0.075]$). On the allowance family C-PR cannot be told apart from zero at any λ and lies far below its allowance layer ($\Delta\tau = -0.356$, interval $[-0.388, -0.323]$) and below EndorseRank (-0.375 , $[-0.404, -0.344]$). The cause is the sparsity of the allowance layer. The transfer layer has about nine times the allowance layer's edges, and 30,791 nodes against 6,442 (Table 4.1), so most nodes of the union graph have no allowance out-edge. At those nodes the mixing rule of Equation (3.9) falls back on the transfer row whatever λ is; λ acts only at the minority of nodes with out-edges in both layers. Once the walk leaves that minority, it travels on transfer edges until it teleports. Raising λ to 0.75 shifts the allowance coefficient by less than one hundredth. The mixture recovers the allowance construct only at the end point $\lambda = 1$; at $\lambda = 0.75$ its allowance coefficient is still 0.006.

Only on the Sybil-adjusted stability family does C-PR exceed both layers walked alone, as the secondary criterion of Section 3.7 asked. It scores 0.290, against 0.286 for C-PR ($\lambda = 0$) and 0.233 for C-PR ($\lambda = 1$), and both paired intervals lie above zero ($[0.003, 0.006]$ and $[0.043, 0.069]$). It also exceeds AWP (0.278) and EndorseRank (0.231) on this family. The gain over the transfer walk is four thousandths, which makes it a difference in the statistical sense but not in any operational one.

On every family, S-PR does worse than the layer it borrows from. Restarting the transfer walk from the allowance walk drops the transfer coefficient to 0.494 ($\Delta\tau = -0.040$ against C-PR ($\lambda = 0$)), raises the allowance coefficient only as far as 0.116 ($\Delta\tau = -0.239$ against C-PR ($\lambda = 1$)), and pushes Sybil-adjusted stability below both layers. A restart vector changes where the walk lands after each teleport, but on a graph of this density it does not change where the walk spends its time. This is the same-window counterpart of the finding that Section 4.6 reaches out of window.

As fixed on 14 September, the secondary criterion asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on Sybil-adjusted stability. At $\lambda = 0.5$, C-PR passes two of the three parts and fails the third. Its transfer coefficient (0.527) falls inside the interval of C-PR ($\lambda = 0$) ($[0.522, 0.547]$), and

its Sybil-adjusted stability exceeds both layers. The allowance coefficient (-0.001), however, lies far outside the interval of C-PR ($\lambda = 1$) ($[0.322, 0.384]$), so the secondary rule fails. The paired contrasts of Table 4.4, reported alongside the rule as a stricter check, would fail the transfer part as well, since they place C-PR slightly but significantly below its transfer layer. In the same-window tables the denser layer dominates the mixture. The holdout takes up the remaining question of whether the mixture has any predictive content that the single layers lack.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same wallets in different orders. The inter-method Kendall τ_b between the two score vectors is 0.360, with interval $[0.342, 0.379]$ (last row of Table 4.5), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Agreement runs above chance mainly because the wallets that EndorseRank scores zero also sit near the bottom of AWP (see below). Because 67.3% of wallet pairs tie on EndorseRank, only 32.7% of all pairs are ordered by both scores, and 18.6% of those pairs are discordant.

The marginal score distributions are in Figure 4.2, and the EndorseRank distribution is dominated by one tie block. EndorseRank takes only 99 distinct values on the cohort: 4,424 wallets share the score 9.62×10^{-5} that a wallet receiving no allowance gets from teleportation and redistributed dangling mass; 966 score zero because no allowance edge touches them; only 131 lie above the common value. Ties are rare under AWP, at 0.5% of wallet pairs. A marginal distribution cannot show how two scores order the same wallets, so Figure 4.3 plots the two rankings against each other, with tied wallets at their mid-rank as in τ_b . The EndorseRank tie blocks therefore appear as vertical columns. The 131 wallets that EndorseRank separates are spread over most of the AWP ranking (median AWP rank 3,096, interquartile range 1,520–4,850), so its leaders are not AWP’s leaders. The column of 4,424 tied wallets spans almost the whole AWP range (interquartile range 1,149–3,571). The 966 zero-score wallets sit mostly near the bottom of AWP (median AWP rank

4,966), and 370 of them score zero under AWP as well.

The tie blocks come from two rules, and one of them is a choice of this pipeline. The 4,424 wallets at the common value are in the allowance graph as owners that receive no approval, so PageRank gives them teleportation and dangling mass and nothing else. The 966 zero-score wallets are not in the graph at all, and the pipeline scores a missing wallet zero (Section 3.6); kept as isolated nodes, they would have received the common value too. This rule decides most of EndorseRank's ordering of the cohort. Of the 4.99 million wallet pairs that EndorseRank does not tie, 4.27 million (85.7%) set a wallet at the common value against a zero-score wallet. On the matched cohort, EndorseRank therefore mostly separates wallets in the allowance graph from wallets outside it. A wallet is in the graph when at least one latest allowance that it granted or received in the window is non-zero. Of the 966 outside it, 802 made no approval in the window; the other 164 did, but on every token and spender the latest approval they made set the allowance to zero.

Nothing in the graph favors the zero rule over the isolated-node alternative, so we reran the same-window alignment with the missing wallets kept as isolated nodes. This check was run after the results above were known. Under uniform teleportation an isolated node scores like any node without in-edges, so the 966 wallets join the block of 4,424 and every other score is multiplied by one common factor. AWP's 381 missing wallets were handled the same way; under AWP's restarts an isolated wallet that has sent nothing receives no restart mass, so it scores zero either way. Table 4.5 gives the result. No AWP coefficient moves. EndorseRank's allowance coefficient rises to 0.990 (interval $[0.988, 0.992]$), since EndorseRank and the allowance proxies now split the cohort at the same place. Its transfer coefficient falls to -0.071 ($[-0.086, -0.056]$), and its Sybil-stability and trading-success coefficients to -0.036 and -0.017 . The agreement between the two scores falls from 0.360 to -0.025 ($[-0.044, -0.003]$). EndorseRank's positive same-window coefficients on transfer, stability and trading, and its agreement with AWP, therefore measure which traders are in the allowance graph.

H1 is supported. The two scores order the cohort differently, and they are related mainly through the wallets that EndorseRank scores zero. What each score carries beyond its ties shows out of window,

Table 4.5: Same-window family-mean Kendall τ with wallets missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 5,521$). The main analysis scores a wallet that is missing from a method’s graph zero. The rule concerns 966 wallets for EndorseRank and 381 for AWP. Under EndorseRank’s uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated wallet keeps the score zero and AWP’s coefficients are the same under both rules. The last row is the agreement between the two scores. The check was run after the results were known. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Family	Missing wallets scored zero		Missing wallets as isolated nodes	
	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.330 [0.311, 0.350]	0.612 [0.602, 0.622]	-0.071 [-0.086, -0.056]	0.612 [0.602, 0.622]
Allowance	0.375 [0.342, 0.401]	-0.024 [-0.043, -0.002]	0.990 [0.988, 0.992]	-0.024 [-0.043, -0.002]
Sybil stability	0.231 [0.212, 0.250]	0.278 [0.264, 0.292]	-0.036 [-0.049, -0.022]	0.278 [0.264, 0.292]
Liquidation	0.093 [0.078, 0.110]	0.141 [0.129, 0.155]	0.082 [0.069, 0.096]	0.141 [0.129, 0.155]
Inverse risk	-0.055 [-0.071, -0.039]	-0.127 [-0.140, -0.115]	0.006 [-0.015, 0.023]	-0.127 [-0.140, -0.115]
GMX trading success	0.096 [0.080, 0.113]	0.236 [0.223, 0.250]	-0.017 [-0.033, -0.003]	0.236 [0.223, 0.250]
EndorseRank vs. AWP	0.360 [0.342, 0.379]		-0.025 [-0.044, -0.003]	

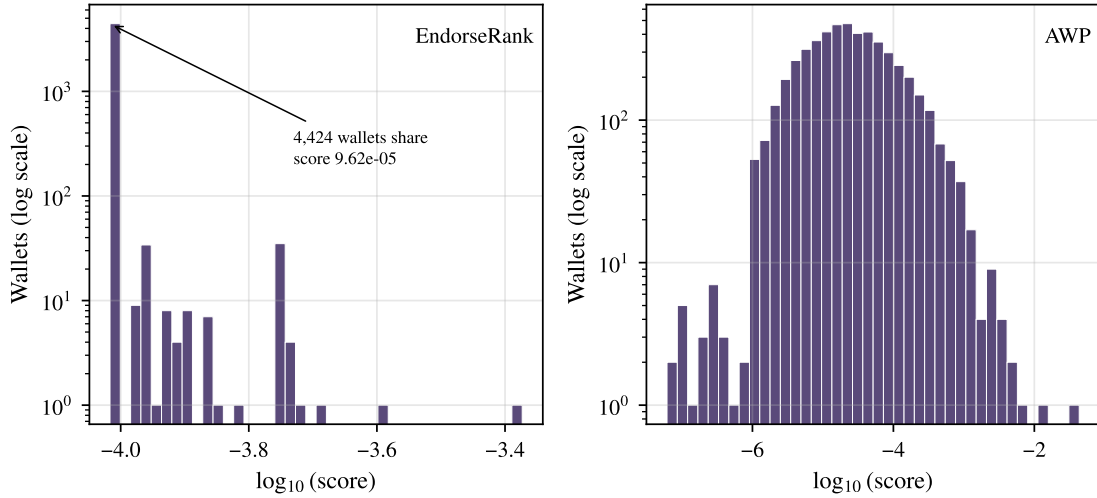


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 5,521$), drawn on a $\log_{10}$ scale for wallets with a positive score; counts are on a log scale. Zero-score wallets, which no edge of the respective graph touches, are left out: 966 for EndorseRank and 381 for AWP. For EndorseRank, 4,424 wallets share the score 9.62×10^{-5} of a wallet that receives no allowance, and only 131 lie above it.

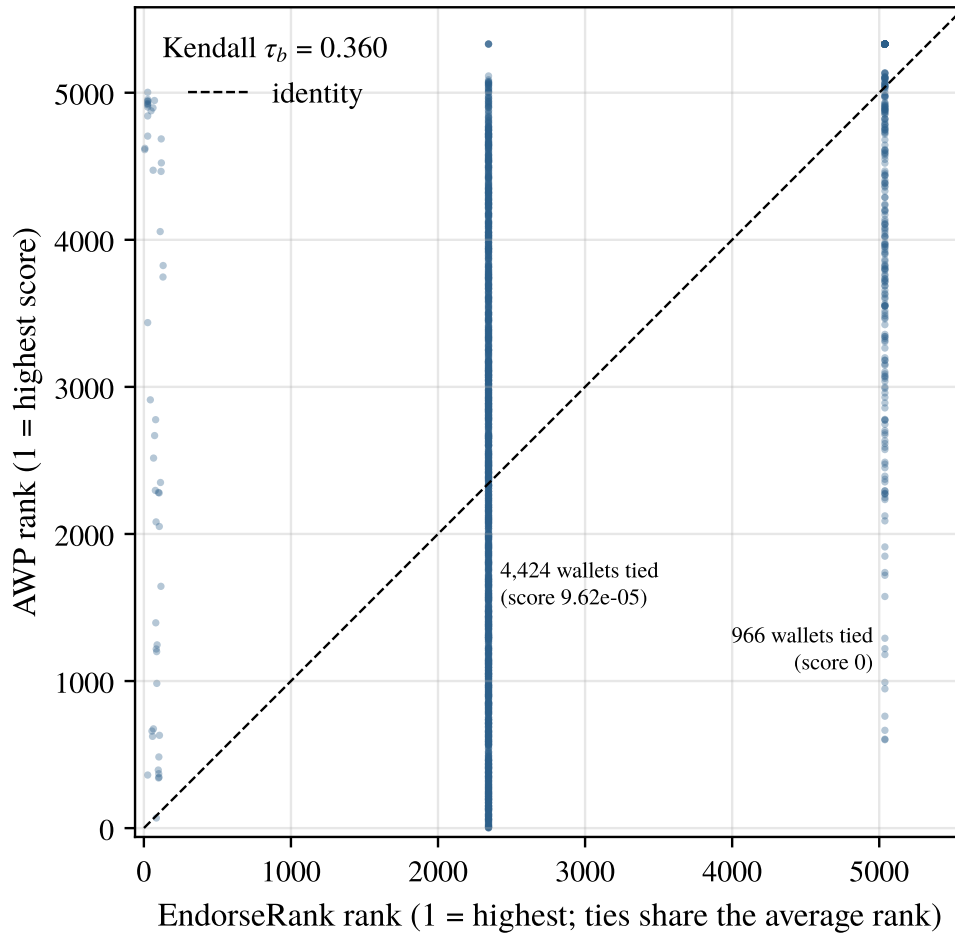


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 2,000 wallets). Tied scores share their average rank, which is the tie treatment of Kendall’s τ_b . The two vertical columns are EndorseRank tie blocks: the 4,424 wallets with the common score 9.62×10^{-5} and the 966 wallets with score zero. A point on the dashed identity line has the same rank in both orderings.

where every spender has received approvals and EndorseRank leaves almost no ties (Section 4.6). A protocol that relies on transfer-based scores cannot assume that its wallet ordering survives a switch to allowance-based scores or a blend with them.

4.5 Robustness and sensitivity checks

Three checks ask whether the results hinge on one damping value, one token mix or one sample definition; they were run for EndorseRank and AWP. The first two hold the sample fixed and perturb a parameter or the token set. The third varies the sample definition itself, and there the coefficients are expected to move. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.6: Robustness: mean Kendall τ under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched alignment cohort ($n = 5,521$). The $d = 0.85$ runtimes are the same measurements as Table 4.1 (one timing session; each graph–damping pair is timed once).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)
0.75	0.375	-0.022	0.330	0.626	0.541	5.011
0.85	0.375	-0.024	0.330	0.612	0.547	5.029
0.95	0.375	-0.030	0.330	0.585	0.541	5.187

For EndorseRank the family means do not move at the third decimal place across $d \in \{0.75, 0.85, 0.95\}$. For AWP the transfer coefficient falls from 0.626 to 0.585 as d rises, and the allowance coefficient stays slightly negative. Neither score’s pattern is an artifact of the conventional $d = 0.85$. EndorseRank’s stability is largely structural. Only 131 cohort wallets lie above EndorseRank’s common value (Section 4.4); a change in d moves the common value but leaves the 4,424 wallets that share it tied, so the rank correlations can hardly move. Runtime barely moves either. EndorseRank takes 0.541, 0.547 and 0.541 s at $d = 0.75, 0.85$ and 0.95 , and AWP 5.011, 5.029 and 5.187 s, so the ratio stays between $9.2\times$ and $9.6\times$. Each graph–damping pair was timed once, in one session. Matrix assembly takes most of each call and does not depend

on d , which is why the extra iterations at $d = 0.95$ add so little. Figure 4.4 plots the family means.

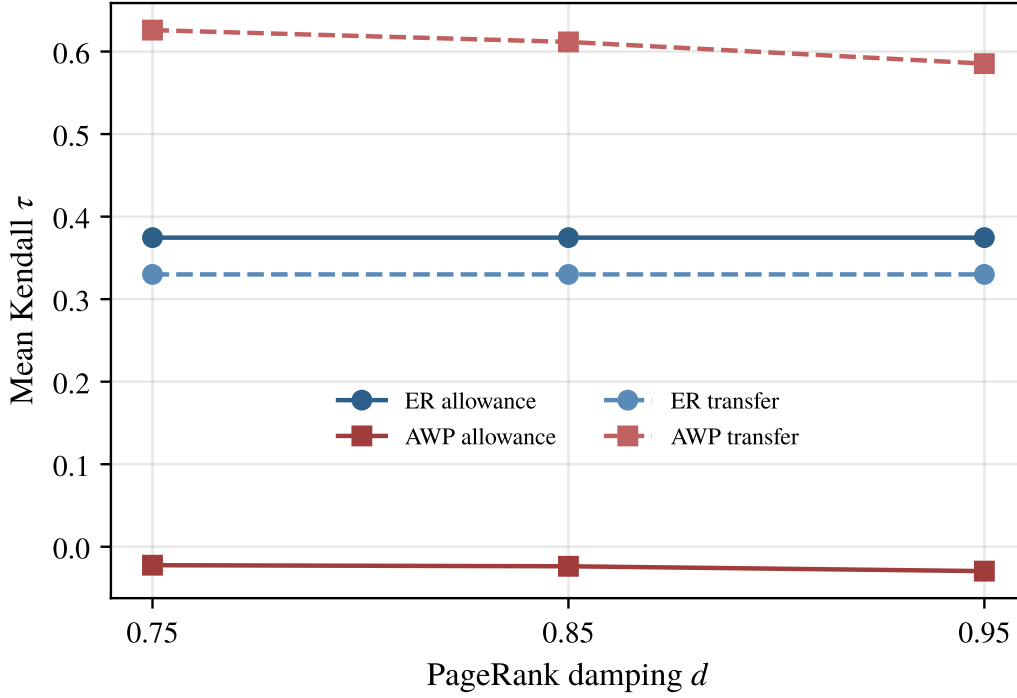


Figure 4.4: Damping sensitivity of family-mean Kendall τ for the allowance and transfer families across $d \in \{0.75, 0.85, 0.95\}$ (values from Table 4.6).

4.5.2 Top-token subgraph

Table 4.7: Robustness: mean Kendall τ on the matched cohort when both graphs keep only 20 ERC-20 tokens, chosen by summed raw amount or by number of rows (transfer events and latest allowances), next to the value on all tokens from Table 4.3.

Method	Transfer			Allowance			Sybil		
	by amount	by count	all	by amount	by count	all	by amount	by count	all
EndorseRank	0.328	0.329	0.330	0.370	0.370	0.375	0.231	0.230	0.231
AWP	0.598	0.603	0.612	-0.021	-0.022	-0.024	0.273	0.275	0.278

When both graphs are restricted to twenty tokens (Table 4.7), the largest change in a family mean is 0.014, for AWP on the transfer family when the tokens are the twenty with the largest summed raw amounts; with the twenty tokens that have the most rows it is 0.009. EndorseRank moves by

0.005 at most. The two lists share eleven tokens, so the pattern does not depend on the many tokens outside either set. Neither list ranks tokens by economic value: summed raw amounts favor tokens with many unlimited approvals, and row counts favor tokens that change hands often.

4.5.3 Sample-definition sensitivity

Table 4.8: Sensitivity of alignment to the evaluation set: mean Kendall τ and runtime at staged subsample sizes drawn from the expanded wallet pool ($N = 99,080$; seed `benchmark-tier2-v1`). Rows are evaluated on different wallet sets from the matched cohort, so τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. Runtimes are the same measurements as Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)	Speedup
10,000	0.343	0.239	0.821	0.900	0.075	1.078	14.37
50,000	0.333	0.230	0.827	0.909	0.342	3.414	9.98
99,080	0.332	0.230	0.826	0.909	0.547	5.029	9.19

In Table 4.8 the transfer and allowance family means are recomputed on the staged pool subsamples from the runtime scaling. The transfer-family means come out much larger than on the matched cohort (0.82–0.91), and AWP’s allowance coefficient turns positive (0.23–0.24), while EndorseRank’s allowance coefficient (0.332–0.343) ends below its cohort value of 0.375. None of these shifts is evidence about the scores. The pool subsamples let in wallets with no edges in one graph or in both; after alignment such a wallet has a zero score and a zero proxy, and tied zeros at the bottom of both rankings push Kendall τ up mechanically (Section 3.12.4). We therefore report the table as a sensitivity analysis of the sample definition, and its values must not be compared with those of Table 4.3. At every pool stage the EndorseRank graph has far fewer nodes than the sample has wallets (Table 4.2), so most pool wallets score zero on it, against 966 of the 5,521 cohort wallets.

The damping and top-token checks show that AWP’s same-window coefficients do not hinge on the parameter or the token mix, and EndorseRank’s could hardly move, because its ties hold whatever d or token set is used. When the population changes, as in the sample-definition check, the absolute

values do not survive.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). Carried over to allowances, AWP’s rule restarts the walk at owners in proportion to the number of spenders they approved recently. Table 4.9 scores both versions of EndorseRank, with AWP beside them. The restart rule was chosen after this comparison had been made.

Table 4.9: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP’s rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 5,521$) and τ with the two spring labels on the spender cohort ($n = 1,335$). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Method	Restarts	Same window, matched cohort			Spring holdout, spenders	
		Transfer	Allowance	Sybil	New approval pairs	New senders
EndorseRank	uniform	0.330 [0.311, 0.350]	0.375 [0.342, 0.401]	0.231 [0.212, 0.250]	0.394 [0.360, 0.431]	0.283 [0.247, 0.320]
EndorseRank	by approving activity	0.445 [0.433, 0.457]	0.044 [0.029, 0.062]	0.245 [0.233, 0.259]	0.372 [0.327, 0.417]	0.301 [0.254, 0.348]
AWP	by sending activity	0.612 [0.602, 0.622]	-0.024 [-0.043, -0.002]	0.278 [0.264, 0.292]	0.123 [0.068, 0.177]	0.344 [0.307, 0.383]

With AWP’s restarts, EndorseRank’s same-window allowance coefficient falls from 0.375 to 0.044, and its transfer coefficient rises from 0.330 to 0.445. The restart mass then goes to the owners that approve most. The cohort’s traders are such owners, so the score orders them by how much they approve, which is closer to their activity than to the endorsements they receive. Out of window the two versions are close. On the spring spenders they reach 0.394 and 0.372 on new approval pairs and 0.283 and 0.301 on new senders, with overlapping intervals, and in June–August 0.333 and 0.325, and 0.255 and 0.284 (Table 4.16). The uniform rule keeps the allowance construct in the same window and gives up nothing measurable out of window.

4.6 Temporal holdout (RQ4, RQ5)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the spender cohort are frozen at 28 February 2026, and two labels are then counted for each wallet between 1 March and 31 May 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.10 sets every score computed at t_1 , the two raw-degree baselines included, against both labels. Table 4.11 gives the paired contrasts, first those fixed before the coupled scores were computed and then those computed after the labels were known.

Table 4.10: Temporal holdout on the spender cohort ($n = 1,335$): Kendall τ between scores frozen at 28 February 2026 and two labels counted on the wallet between 1 March 2026 and 31 May 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the wallet for the first time. CI = 95% percentile bootstrap over wallets (400 paired resamples). 222 wallets received at least one new pair and 175 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.711 [0.665, 0.759]	0.404 [0.348, 0.467]
Transfer in-degree at t_1 (baseline)	0.117 [0.049, 0.191]	0.472 [0.423, 0.520]
EndorseRank	0.394 [0.360, 0.431]	0.283 [0.247, 0.320]
AWP	0.123 [0.068, 0.177]	0.344 [0.307, 0.383]
C-PR ($\lambda = 1$)	0.479 [0.427, 0.529]	0.372 [0.325, 0.424]
C-PR ($\lambda = 0$)	0.044 [-0.011, 0.098]	0.270 [0.235, 0.310]
C-PR ($\lambda = 0.25$)	0.271 [0.231, 0.314]	0.266 [0.231, 0.306]
C-PR ($\lambda = 0.5$)	0.295 [0.252, 0.336]	0.277 [0.241, 0.318]
C-PR ($\lambda = 0.75$)	0.314 [0.271, 0.357]	0.287 [0.253, 0.330]
S-PR	0.114 [0.052, 0.164]	0.264 [0.229, 0.304]

We take the two findings in turn.

The first finding answers the second part of RQ4 in the negative: no PageRank improves on the raw degree of its own edge type at t_1 . In-approve degree at t_1 predicts new approval pairs with $\tau = 0.711$. The walks over allowances come closest and still fall far short: C-PR ($\lambda = 1$) reaches 0.479,

Table 4.11: Paired bootstrap differences on the spender holdout ($n = 1,335,400$ resamples shared by a and b). The first block was fixed before the run: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1 , and the hybrid contrasts fixed on 14 September 2026, which compare C-PR and S-PR with the two layers they are built from, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). The second block was computed after the labels were known: the increments of EndorseRank and AWP over their raw degrees, C-PR against C-PR ($\lambda = 0$) on new approval pairs, and C-PR against EndorseRank and AWP. Primary criterion of 14 September (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): not met. The differences between EndorseRank and AWP are in Table 8.8.

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
<i>Fixed before the run</i>						
C-PR ($\lambda = 1$) minus t1 in-approve degree	new approval pairs	0.479	0.711	-0.233	[-0.291, -0.183]	0.000
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.295	0.479	-0.184	[-0.213, -0.149]	0.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.277	0.270	+0.007	[-0.012, 0.028]	0.735
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.114	0.479	-0.365	[-0.431, -0.300]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.264	0.270	-0.006	[-0.017, 0.006]	0.163
<i>Computed after the labels were known</i>						
EndorseRank minus t1 in-approve degree	new approval pairs	0.394	0.711	-0.317	[-0.354, -0.283]	0.000
AWP minus t1 in-degree	new transfer senders	0.344	0.472	-0.127	[-0.150, -0.099]	0.000
C-PR minus C-PR ($\lambda = 0$)	new approval pairs	0.295	0.044	+0.251	[0.203, 0.306]	1.000
C-PR minus EndorseRank	new approval pairs	0.295	0.394	-0.099	[-0.133, -0.064]	0.000
C-PR minus AWP	new transfer senders	0.277	0.344	-0.067	[-0.095, -0.039]	0.000
C-PR minus AWP	new approval pairs	0.295	0.123	+0.172	[0.111, 0.226]	1.000

a paired difference of -0.233 (interval $[-0.291, -0.183]$, fixed before the run), and EndorseRank 0.394 , a difference of -0.317 ($[-0.354, -0.283]$). Transfer in-degree at t_1 predicts new transfer senders with $\tau = 0.472$, against 0.344 for AWP ($\Delta\tau = -0.127$, interval $[-0.150, -0.099]$) and 0.270 for C-PR ($\lambda = 0$). Every interval lies well below zero. Within a three-month window, a wallet gains more new counterparties the more it already has, and the local count captures this directly. PageRank replaces the count with a weighted, propagated quantity. In EndorseRank an owner's endorsement is shared among all the spenders it approves, and in AWP a sender's among all the addresses it pays; on these labels that sharing acts as noise. If a PageRank adds anything beyond its degree, it does not show in predicting the future growth of its own construct.

Read across the table, the raw degree of each label's own edge type ranks that label best. On new approval pairs the two walks over allowances come next, then the interior mixtures (0.271 to 0.314), while the transfer in-degree, AWP, C-PR ($\lambda = 0$) and S-PR stay at or below 0.123 . On new transfer senders the order is less clean. The in-approve degree (0.404) and C-PR ($\lambda = 1$) (0.372) rank this label better than any walk over transfers, of which AWP reaches the most (0.344); on this cohort the spenders that many owners approve also gain many new senders. EndorseRank leaves almost no ties here: it gives $1,291$ distinct scores to the $1,335$ spenders, and its largest tie block holds 42 of them, so its coefficients rest on an ordering of nearly every spender. C-PR ($\lambda = 1$), the allowance layer walked with raw amounts, ranks both labels higher than EndorseRank. Its rows are decided by unlimited approvals (Section 3.6.3), so on this cohort the raw amounts kept part of the signal that counting each approval once gives up; the thesis does not test why.

Coverage does not explain why the walks fall short of their degrees. Of the $1,335$ spenders, 318 have no transfer edge up to the freeze and score zero under AWP. They gained new approval pairs more often than the rest (22% against 15%) and new senders less often (4% against 16%), so AWP's zeros lower its coefficient on the first label and raise it on the second. Table 4.12 restricts the holdout to the 977 spenders that had received a transfer from another address before the freeze. There AWP stays far below the transfer in-degree (0.388 against 0.701 ; -0.314 , $[-0.353, -0.276]$), and EndorseRank stays below the in-approve degree (0.340 against 0.679). The restriction was chosen

after the results were known.

Table 4.12: Spender holdout restricted to the 977 of the 1,335 spenders that had received a transfer from another address before the freeze. The other 358 have no inbound transfer edge, and 318 spenders score zero under AWP. The restriction was tried after the results were known. The differences between EndorseRank and AWP on this subset are in Table 8.8. CI = 95% percentile bootstrap over wallets (400 resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.340 [0.299, 0.386]	0.358 [0.316, 0.396]
AWP	0.298 [0.252, 0.343]	0.388 [0.348, 0.430]
In-approve degree at t_1	0.679 [0.616, 0.736]	0.570 [0.505, 0.635]
Transfer in-degree at t_1	0.569 [0.510, 0.631]	0.701 [0.651, 0.750]
AWP minus t1 in-degree (new transfer senders)	$\Delta\tau = -0.314$ [-0.353, -0.276]	

The second finding is that the coupled operator fails the rule of 14 September, although in an exploratory comparison it improves on its transfer layer walked alone. That rule asked C-PR to match C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders. At $\lambda = 0.5$, C-PR does worse than the allowance walk on new approval pairs ($\Delta\tau = -0.184$, interval $[-0.213, -0.149]$) and no differently from the transfer walk on new transfer senders ($\Delta\tau = +0.007$, interval $[-0.012, 0.028]$), so both conditions of its primary part fail. Against the transfer walk alone, a comparison the rule did not include, C-PR does better. On new approval pairs C-PR reaches 0.295 against 0.044, a paired difference of +0.251 (interval $[0.203, 0.306]$, on the same resamples), and on new senders it shows no detectable loss. This contrast was computed after the labels were known, so it is treated as a hypothesis and tested again in the registered window below. Against EndorseRank and AWP, also computed after the labels were known, C-PR sits between the two on new approval pairs, 0.099 below EndorseRank ($[-0.133, -0.064]$) and 0.172 above AWP ($[0.111, 0.226]$), and it falls behind AWP on new senders ($-0.067, [-0.095, -0.039]$). On the spring data the analogue of F2 would therefore fail against AWP. The registered replication reports that comparison next to its decision (Section 3.7.2). Across the three values of λ the order is monotone on both labels. On new approval pairs τ rises from 0.271 ($\lambda = 0.25$) through 0.295 to 0.314 ($\lambda = 0.75$), and on new transfer senders from 0.266 through 0.277 to 0.287. More allowance weight helps on both labels;

for this cohort the transfer layer adds no out-of-window signal to what the allowance layer carries. The seeded ablation shows where the allowance signal has to sit. S-PR matches the transfer walk on the flow label ($\Delta\tau = -0.006$, interval $[-0.017, 0.006]$) and falls far below the allowance walk on the endorsement label ($\Delta\tau = -0.365$). When transfer edges decide where the walk goes, a restart distribution derived from endorsements changes very little. On the spender holdout, the allowance signal survives when allowance edges carry the walk, fully in the allowance walk and diluted in C-PR, and it is lost when they only decide where the walk restarts.

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these results bears on credit or loss. Both labels are also sparse. Of the 1,335 wallets, 222 gained a new approval pair and 175 a new sender; every coefficient rests entirely on how that small subset is ordered.

Other candidate labels were computed on the holdout, and they are reported here without being offered as support. Revocation counts and the drain heuristic both rise with EndorseRank ($\tau = 0.269$, interval $[0.226, 0.308]$, and 0.422 , interval $[0.389, 0.455]$), and more steeply still with the raw in-approve degree (0.483 and 0.575). For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. The likelier reading is volume. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use. Aave liquidations are empty in this spender set.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 was scored after its labels were known and is reported as exploratory. Table 4.13 gives every score on the GMX labels of the 1,860 matched wallets with at least three closes from March to May.

The scores built on transfers rank the number of profitable closes and the realized gain. AWP reaches 0.210 and 0.191 , the transfer in-degree 0.198 and 0.169 , and the walks over the raw transfer layer and its mixtures 0.111 to 0.141 and 0.120 to 0.154 . EndorseRank and C-PR ($\lambda = 1$) reach about 0.07 and 0.05 , and the in-approve degree nothing. Both labels grow with how much a wallet

Table 4.13: Spring trader run: Kendall τ between scores frozen at 28 February 2026 and GMX V2 labels counted from 1 March 2026 to 31 May 2026, on the 1,860 matched wallets with at least three closes in that window. The run was scored after its labels were known. Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	-0.008 [-0.038, 0.026]	0.013 [-0.018, 0.049]	-0.057 [-0.095, -0.015]	-0.102 [-0.138, -0.063]
Transfer in-degree at t_1 (baseline)	0.198 [0.168, 0.228]	0.169 [0.135, 0.199]	0.016 [-0.017, 0.048]	0.011 [-0.017, 0.041]
EndorseRank	0.072 [0.041, 0.111]	0.048 [0.017, 0.082]	-0.016 [-0.053, 0.024]	-0.016 [-0.054, 0.023]
AWP	0.210 [0.181, 0.238]	0.191 [0.159, 0.219]	0.016 [-0.015, 0.044]	-0.020 [-0.051, 0.006]
C-PR ($\lambda = 1$)	0.073 [0.043, 0.111]	0.048 [0.018, 0.082]	-0.015 [-0.051, 0.027]	-0.016 [-0.054, 0.023]
C-PR ($\lambda = 0$)	0.131 [0.101, 0.162]	0.153 [0.120, 0.182]	0.029 [-0.003, 0.058]	0.012 [-0.014, 0.045]
C-PR ($\lambda = 0.25$)	0.128 [0.096, 0.158]	0.154 [0.122, 0.184]	0.025 [-0.007, 0.055]	0.007 [-0.019, 0.038]
C-PR ($\lambda = 0.5$)	0.121 [0.089, 0.152]	0.151 [0.120, 0.183]	0.024 [-0.009, 0.053]	0.007 [-0.019, 0.039]
C-PR ($\lambda = 0.75$)	0.111 [0.079, 0.140]	0.146 [0.114, 0.177]	0.023 [-0.009, 0.052]	0.009 [-0.019, 0.041]
S-PR	0.141 [0.114, 0.173]	0.120 [0.089, 0.150]	0.014 [-0.016, 0.047]	-0.008 [-0.034, 0.021]

trades. On the two shares, which do not, no coefficient exceeds 0.03, and the in-approve degree is negative on both. That is what one should expect from a transfer score, which tracks activity, and from an allowance score that leaves most traders tied; none of the scores ranks how well a wallet trades. The Aave liquidation label of the same run covers only 174 wallets, 10 of them liquidated, which is too few to read. The registered window gives C-PR a trader label of its own.

4.6.2 Registered replication: June to August 2026

The registration of Section 3.7.2 was committed to the public repository on 1 October 2026, and the June–August logs were extracted later that day. Nine monthly queries over approvals, transfers and GMX events scanned about 1.04 TB and returned 163,553 approval logs, 1,013,023 transfer logs and 144,152 GMX closes, all of which decoded. The extraction manifest records the hashes of both registration files and no use of the testing override. Before any fresh number was computed, the evaluation code was run on the spring window as a check. It reproduced exactly the sixteen spring coefficients of the eight scores it shares with Table 4.10 (the two layers walked alone, C-PR at three values of λ , S-PR and the two degrees) and the contrasts of the 14 September rule, and it gave the difference of +0.251 quoted above for C-PR against its transfer layer on new approval pairs. The registered evaluation then ran with 400 paired resamples and seed 42.

At the freeze on 31 May, 1,964 spenders held a positive latest allowance; 158 of them gained a new approval pair from June to August and 136 a new sender. Of the 5,521 matched wallets, 5,151 were in the freeze-date graph, and 1,402 of these closed at least three positions in the label window. Of the 1,402, 789 had at least one liquidation and 17 had every close liquidated. Table 4.14 gives every registered score on the three registered labels, and Table 4.15 gives the contrasts with their verdicts. The registration documents call the two layers walked alone EndorseRank and AWP; the tables use the names of this thesis.

The rule is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.238$ against 0.058 for its transfer layer walked alone, a difference of +0.179 (interval [0.132, 0.225]), so the spring gain on authorization replicates. F3 holds as well. On the liquidation-

Table 4.14: Registered replication: Kendall τ between scores frozen at 31 May 2026 and the three registered labels, counted from 1 June to 31 August 2026. The spenders are every spender holding a positive latest allowance at the freeze; 158 of them gained a new approval pair and 136 a new sender. The traders are the 5,151 matched wallets in the freeze-date graph; the label is defined for the 1,402 with at least three closes in the label window, 789 of which had at least one liquidation. C-PR ($\lambda = 0$) is the comparator of the registered rule and AWP that of a registered sensitivity analysis (Table 4.15). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at 31 May 2026	Spenders ($n = 1,964$)		Traders ($n = 1,402$)
	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.613 [0.564, 0.663]	0.371 [0.313, 0.429]	0.144 [0.110, 0.172]
Transfer in-degree at the freeze (baseline)	0.128 [0.064, 0.188]	0.435 [0.401, 0.468]	-0.006 [-0.046, 0.031]
AWP	0.118 [0.071, 0.161]	0.319 [0.291, 0.345]	0.045 [0.009, 0.079]
C-PR ($\lambda = 1$)	0.338 [0.292, 0.388]	0.289 [0.247, 0.332]	0.086 [0.037, 0.129]
C-PR ($\lambda = 0$)	0.058 [0.012, 0.103]	0.257 [0.229, 0.286]	0.065 [0.030, 0.102]
C-PR ($\lambda = 0.25$)	0.228 [0.190, 0.265]	0.245 [0.217, 0.273]	0.077 [0.042, 0.114]
C-PR ($\lambda = 0.5$)	0.238 [0.200, 0.274]	0.246 [0.218, 0.274]	0.083 [0.047, 0.119]
C-PR ($\lambda = 0.75$)	0.243 [0.206, 0.280]	0.248 [0.219, 0.275]	0.086 [0.051, 0.121]
S-PR	0.126 [0.076, 0.172]	0.254 [0.223, 0.281]	0.030 [-0.006, 0.066]

free close rate of the traders C-PR reaches 0.083 against 0.065 for the transfer walk (+0.018, [0.011, 0.024]). F2 fails. On new transfer senders C-PR reaches 0.246 against 0.257; the difference is -0.011 , and the lower end of its interval, -0.025 , lies below the margin of -0.02 . The interval also contains zero, so the data do not show that C-PR is worse than the transfer walk on new senders. Nor do they show that it is within 0.02 of it, which is what the rule asked. The rule needed all three hypotheses, so, as the plan said for this case, the thesis makes no claim that C-PR improves on a walk over the transfers alone.

The plan warned that F3 could pass with both coefficients close to zero. Both are small, and both intervals lie above zero (C-PR [0.047, 0.119], the transfer walk [0.030, 0.102]), so each walk carries a little information about which traders avoid liquidation. The in-approve degree at the freeze ranks this label better than any walk (0.144, [0.110, 0.172]); the allowance walk reaches 0.086 and AWP 0.045.

The results reported beside the decision follow the same order. F4 holds: on the liquidation-free close rate C-PR is better than the transfer walk, and not only no worse. F5 fails, since C-PR and

Table 4.15: Registered replication: the paired contrasts F1–F6 with their verdicts, and the two sensitivity analyses added before the extraction. $\Delta\tau = \tau_a - \tau_b$, with C-PR as a . The registration documents call the two layers walked alone, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), EndorseRank and AWP. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above -0.02 ; F6a, the first part of the 14 September rule, holds when the interval contains zero or lies above it. F1–F3 are on spenders (new approval pairs, new transfer senders) and traders (liquidation-free close rate) as in Table 4.14. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast	Label	Test	τ_a	τ_b	$\Delta\tau$ [95% CI]	Verdict
<i>Registered rule: C-PR ($\lambda = 0.5$) against its transfer layer alone, C-PR ($\lambda = 0$)</i>						
F1	C-PR vs. C-PR ($\lambda = 0$) new approval pairs	superiority	0.238	0.058	+0.179 [0.132, 0.225]	holds
F2	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	non-inferiority, -0.02	0.246	0.257	-0.011 [-0.025, 0.003]	fails
F3	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	non-inferiority, -0.02	0.083	0.065	+0.018 [0.011, 0.024]	holds
Decision: F1, F2 and F3 must all hold						not met
<i>Reported, not part of the decision</i>						
F4	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	superiority	0.083	0.065	+0.018 [0.011, 0.024]	holds
F5	C-PR vs. C-PR ($\lambda = 1$) liquidation-free close rate	superiority	0.083	0.086	-0.003 [-0.060, 0.051]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$) new approval pairs	no worse	0.238	0.338	-0.101 [-0.127, -0.076]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	superiority	0.246	0.257	-0.011 [-0.025, 0.003]	fails
<i>Sensitivity analysis: AWP as the comparator</i>						
F1	C-PR vs. AWP new approval pairs	superiority	0.238	0.118	+0.119 [0.075, 0.169]	holds
F2	C-PR vs. AWP new transfer senders	non-inferiority, -0.02	0.246	0.319	-0.073 [-0.092, -0.055]	fails
F3	C-PR vs. AWP liquidation-free close rate	non-inferiority, -0.02	0.083	0.045	+0.037 [0.011, 0.066]	holds
F4	C-PR vs. AWP liquidation-free close rate	superiority	0.083	0.045	+0.037 [0.011, 0.066]	holds
<i>Sensitivity analysis: wallets outside a graph kept as isolated nodes</i>						
F1	C-PR vs. C-PR ($\lambda = 0$) new approval pairs	superiority	0.238	0.053	+0.185 [0.137, 0.232]	holds
F2	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	non-inferiority, -0.02	0.246	0.258	-0.012 [-0.027, 0.003]	fails
F3	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	non-inferiority, -0.02	0.083	0.065	+0.018 [0.010, 0.024]	holds
F4	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	superiority	0.083	0.065	+0.018 [0.010, 0.024]	holds
F5	C-PR vs. C-PR ($\lambda = 1$) liquidation-free close rate	superiority	0.083	0.155	-0.072 [-0.118, -0.028]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$) new approval pairs	no worse	0.238	0.338	-0.101 [-0.127, -0.076]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	superiority	0.246	0.258	-0.012 [-0.027, 0.003]	fails

the allowance walk cannot be told apart on that label ($-0.003, [-0.060, 0.051]$). F6, the primary part of the 14 September rule, fails again on both parts: C-PR is below the allowance walk on new approval pairs ($-0.101, [-0.127, -0.076]$) and not above the transfer walk on new senders. C-PR's coefficients rise with λ on both spender labels, as in the spring. S-PR matches the transfer walk on new senders ($-0.003, [-0.017, 0.010]$) and falls far below the allowance walk on new approval pairs (-0.213).

Neither registered sensitivity analysis changes the verdict (Table 4.15). With AWP as the comparator, F1 and F3 still hold ($+0.119$ and $+0.037$) and F2 fails clearly. AWP ranks new senders at 0.319, and C-PR's difference to it is -0.073 ($[-0.092, -0.055]$), an interval that lies entirely below the margin. Against AWP, then, C-PR gives up a measurable part of the ranking of later inflow in exchange for its gain on later approvals. With the wallets outside a graph kept as isolated nodes, F1 to F4 keep their verdicts. F5 changes from a null to a loss. The rule moves the matched traders outside the allowance graph from the bottom of the allowance walk's order into the block of wallets without in-edges (Section 4.4); the allowance walk's coefficient on the liquidation-free close rate then rises from 0.086 to 0.155, and C-PR falls below it ($-0.072, [-0.118, -0.028]$). F6 fails under this rule too.

Table 4.16: EndorseRank on the cohorts and labels of the registered replication, scored after that replication was evaluated; none of these rows enters its decision. The AWP, C-PR and degree rows repeat Table 4.14, and the contrasts use the same wallet resamples; the differences between EndorseRank and AWP are in Table 8.8. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at 31 May 2026	Spenders ($n = 1,964$)		Traders
	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.613 [0.564, 0.663]	0.371 [0.313, 0.429]	0.144 [0.110, 0.172]
Transfer in-degree at the freeze (baseline)	0.128 [0.064, 0.188]	0.435 [0.401, 0.468]	-0.006 [-0.046, 0.031]
EndorseRank	0.333 [0.306, 0.364]	0.255 [0.227, 0.284]	0.081 [0.032, 0.123]
EndorseRank, restarts by approving activity	0.325 [0.295, 0.358]	0.284 [0.252, 0.312]	0.077 [0.037, 0.116]
AWP	0.118 [0.071, 0.161]	0.319 [0.291, 0.345]	0.045 [0.009, 0.079]
C-PR ($\lambda = 0.5$)	0.238 [0.200, 0.274]	0.246 [0.218, 0.274]	0.083 [0.047, 0.119]
EndorseRank minus t1 in-approve degree, new approval pairs	$\Delta\tau = -0.280$ [-0.313, -0.247]		
AWP minus t1 in-degree, new transfer senders	$\Delta\tau = -0.116$ [-0.133, -0.095]		
C-PR minus EndorseRank, new approval pairs	$\Delta\tau = -0.096$ [-0.128, -0.065]		
C-PR minus EndorseRank, liquidation-free close rate	$\Delta\tau = +0.002$ [-0.053, 0.056]		

EndorseRank was not among the registered scores, so Table 4.16 scores it on the same cohorts, labels and wallet resamples after the registered evaluation; none of its rows enters the decision. The spring pattern holds for every PageRank in this window too. The in-approve degree at the freeze again ranks new approval pairs above every walk (0.613; EndorseRank minus the degree -0.280 , $[-0.313, -0.247]$), and the transfer in-degree again ranks new senders above every walk (0.435; AWP minus the degree -0.116 , $[-0.133, -0.095]$). C-PR again sits below EndorseRank on new approval pairs (-0.096 , $[-0.128, -0.065]$). On the liquidation-free close rate of the traders EndorseRank reaches 0.081, close to C-PR (0.083; $+0.002$, $[-0.053, 0.056]$). With AWP's restarts, EndorseRank gives 0.325 and 0.284 on the two spender labels and 0.077 on the traders.

Table 4.17: Registered replication, traders: Kendall τ between scores frozen at 31 May 2026 and the other GMX V2 labels counted from June to August 2026, on the 1,402 matched wallets with at least three closes in that window. No liquidation is 1 when none of the wallet's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter the decision. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at 31 May 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.187 [0.150, 0.221]	-0.004 [-0.033, 0.028]	0.021 [-0.015, 0.054]	-0.074 [-0.105, -0.042]	-0.139 [-0.172, -0.101]
Transfer in-degree at the freeze (baseline)	-0.072 [-0.115, -0.030]	0.191 [0.150, 0.224]	0.127 [0.089, 0.166]	0.026 [-0.008, 0.060]	-0.001 [-0.036, 0.038]
AWP	-0.030 [-0.069, 0.011]	0.233 [0.193, 0.268]	0.182 [0.142, 0.217]	0.030 [-0.004, 0.060]	-0.035 [-0.065, -0.001]
C-PR ($\lambda = 1$)	0.069 [0.011, 0.116]	0.074 [0.036, 0.107]	0.016 [-0.024, 0.059]	-0.043 [-0.084, -0.000]	-0.130 [-0.166, -0.091]
C-PR ($\lambda = 0$)	0.014 [-0.026, 0.056]	0.140 [0.103, 0.177]	0.145 [0.110, 0.180]	0.033 [-0.001, 0.069]	-0.004 [-0.040, 0.035]
C-PR ($\lambda = 0.25$)	0.032 [-0.007, 0.072]	0.133 [0.094, 0.168]	0.145 [0.110, 0.182]	0.027 [-0.007, 0.062]	-0.011 [-0.047, 0.028]
C-PR ($\lambda = 0.5$)	0.043 [0.004, 0.082]	0.124 [0.084, 0.160]	0.142 [0.107, 0.179]	0.024 [-0.010, 0.058]	-0.010 [-0.045, 0.028]
C-PR ($\lambda = 0.75$)	0.053 [0.011, 0.091]	0.110 [0.068, 0.147]	0.135 [0.101, 0.171]	0.021 [-0.014, 0.056]	-0.008 [-0.041, 0.030]
S-PR	-0.030 [-0.070, 0.014]	0.176 [0.139, 0.211]	0.132 [0.094, 0.170]	0.040 [0.003, 0.075]	-0.023 [-0.057, 0.015]

Table 4.17 gives the registered trader labels that do not enter the decision, and they repeat the spring trader run. AWP ranks the number of profitable closes (0.233) and the realized gain (0.182) highest

of the registered scores; the transfer in-degree reaches 0.191 and 0.127, and C-PR 0.124 and 0.142. Every score stays at or below 0.04 on the share of closes that were profitable and at or below zero on the share that were not losses.

4.7 Primary claims (C1–C5)

The chapter’s results support the main claims below, cross-referenced to the hypothesis and research question columns of Table 1.2.

- C1. On identical wallet sets, the cost of each score’s solver call follows the number of edges in its graph, because both matrix assembly and iteration scale with $|E|$. EndorseRank’s graph has about one-ninth of the transfer graph’s edges, and its call takes a matching fraction of AWP’s time and memory (Table 4.1). The ratio holds across the in-cohort and pool stages (Tables 4.2 and 8.1), although the pool stages do not test edge counts beyond those of the cohort graph. The low cost belongs to EndorseRank alone: C-PR and S-PR walk the transfer layer and take longer than AWP.
- C2. In the same window each score agrees with the proxies of the edge type it walks on, with intervals that exclude zero, and the mixtures follow their denser transfer layer (Table 4.3). These tests measure internal consistency, and part of the agreement is built in; external validity is outside their scope. EndorseRank’s coefficients are set by the rule for wallets outside the allowance graph (Table 4.5).
- C3. EndorseRank and AWP order the cohort differently. Their agreement has an interval that excludes zero and one, comes mainly from the wallets outside the allowance graph, and falls slightly below zero when those wallets are kept as isolated nodes (Table 4.5, Figure 4.3).
- C4. Out of window no PageRank of this thesis ranks a spender label as well as the raw degree of the label’s own edge type, in either window: every paired difference to the degree lies below zero (Tables 4.11 and 4.16), and the shortfall is not a matter of coverage (Table 4.12). On the matched traders the scores built on transfers rank counts of later profitable closes, and no

score ranks the share of closes that were profitable (Table 4.13). Apart from the increment of C-PR ($\lambda = 1$), none of these comparisons was fixed in advance.

- C5. A single random walk over both edge types does not beat its two layers walked alone, and the registered replication does not confirm that it improves on the transfer walk. Under the rule of 14 September, C-PR falls between the two walks on new approval pairs at every λ tested, and S-PR reproduces the transfer walk (Tables 4.10 and 4.11). The spring gain over the transfer walk on new approval pairs, found after the labels were known, replicates in the registered window (F1, with F3 and F4), but the difference on new senders cannot be bounded within 0.02 (F2), so the registered rule is not met. Against AWP, in the registered sensitivity analysis, C-PR loses clearly on new senders (Table 4.15). Section 4.3.1 reports where the hybrids stand within the window.

The rank divergence answers RQ1, the same-window tables RQ2, the benchmark tables RQ3, and the holdouts RQ4 and RQ5. The direct comparisons between EndorseRank and AWP are in Appendix 8.5, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the research questions and objectives of Chapter 1.